Answer key

SAT Practice Test 4, Reading and Writing

Module 1

NO.	ANSWER	SKILL
1	**A**	Words in context
2	**D**	Words in context
3	**C**	Words in context
4	**B**	Words in context
5	**A**	Words in context
6	**D**	Text structure and purpose
7	**C**	Text structure and purpose
8	**B**	Text structure and purpose
9	**B**	Cross-text connections
10	**A**	Central ideas and details
11	**D**	Central ideas and details
12	**C**	Command of evidence: quantitative
13	**D**	Command of evidence: quantitative
14	**A**	Command of evidence: textual
15	**B**	Command of evidence: textual
16	**C**	Command of evidence: textual
17	**C**	Inferences
18	**A**	Inferences
19	**D**	Inferences
20	**B**	Form, structure, and sense
21	**A**	Boundaries
22	**C**	Boundaries
23	**D**	Form, structure, and sense
24	**B**	Form, structure, and sense
25	**A**	Form, structure, and sense
26	**C**	Form, structure, and sense
27	**B**	Boundaries
28	**D**	Transitions
29	**D**	Transitions
30	**B**	Transitions
31	**A**	Rhetorical synthesis
32	**C**	Rhetorical synthesis
33	**C**	Rhetorical synthesis

Module 2

NO.	ANSWER	SKILL
1	**C**	Words in context
2	**D**	Words in context
3	**B**	Words in context
4	**A**	Words in context
5	**C**	Words in context
6	**A**	Words in context
7	**D**	Text structure and purpose
8	**B**	Text structure and purpose
9	**C**	Text structure and purpose
10	**B**	Cross-text connections
11	**A**	Central ideas and details
12	**D**	Central ideas and details
13	**C**	Command of evidence: quantitative
14	**D**	Command of evidence: quantitative
15	**B**	Command of evidence: textual
16	**A**	Command of evidence: textual
17	**C**	Command of evidence: textual
18	**B**	Inferences
19	**D**	Inferences
20	**A**	Inferences
21	**B**	Boundaries
22	**C**	Boundaries
23	**A**	Form, structure, and sense
24	**D**	Form, structure, and sense
25	**B**	Form, structure, and sense
26	**C**	Boundaries
27	**D**	Form, structure, and sense
28	**A**	Transitions
29	**C**	Transitions
30	**D**	Transitions
31	**A**	Rhetorical synthesis
32	**B**	Rhetorical synthesis
33	**A**	Rhetorical synthesis

Answer key

SAT Practice Test 4, Math

Module 1

NO.	ANSWER	SKILL
1	**B**	Percentages
2	**D**	Data analysis
3	**C**	Linear equations
4	**A**	Linear equations in context
5	**A**	Systems of equations
6	**D**	Functions
7	**B**	Nonlinear expressions
8	**C**	Triangles
9	**D**	Circles
10	**C**	Quadratics
11	**A**	Probability
12	**B**	Statistics
13	**D**	Scatterplots
14	**A**	Inequalities
15	**B**	Functions
16	**C**	Exponential functions
17	**D**	Linear functions
18	**B**	Percent change
19	**C**	Volume
20	**A**	Right triangles
21	**24**	Linear equations
22	**6**	Linear equations
23	**6**	Ratios and rates
24	**80**	Area
25	**15**	Systems of equations
26	**63**	Volume
27	**3**	Functions

Module 2

NO.	ANSWER	SKILL
1	**C**	Linear functions
2	**A**	Percentages
3	**B**	Probability
4	**D**	Exponential functions
5	**B**	Quadratics
6	**C**	Volume
7	**A**	Inequalities
8	**D**	Exponential functions
9	**D**	Circles
10	**C**	Equivalent expressions
11	**A**	Percentages
12	**B**	Absolute value
13	**B**	Exponential equations
14	**A**	Circles in the plane
15	**D**	Quadratics
16	**C**	Percentages
17	**D**	Functions
18	**B**	Trigonometry
19	**C**	Ratios and rates
20	**A**	Equivalent expressions
21	**26**	Systems of equations
22	**11**	Quadratics
23	**20**	Percentages
24	**14**	Systems of equations
25	**45**	Similar triangles
26	**10**	Quadratics
27	**-5**	Systems with nonlinear equations

Score yourself. Count the questions you got right in each module.

Reading and Writing, Module 1: ___ of 33

Module 2: ___ of 33

Math, Module 1: ___ of 27

Module 2: ___ of 27

Answer explanations

SAT Practice Test 4, Reading and Writing, Module 1

Questions 1 to 17

1 **Choice A** **Words in context**

She doubted her equipment because the results clashed with the standard model, so inconsistent is correct. Compatible would give her no reason to doubt.

2 **Choice D** **Words in context**

A landmark every sailor knew was prominent. Hidden contradicts that, and the passage does not describe danger or impermanence.

3 **Choice C** **Words in context**

A village with only wind and a bell is quiet, so the stillness was soothing. Clamour, congestion and bustle describe noise or crowds.

4 **Choice B** **Words in context**

Saving a third of the cost implies the expensive item was removed. Defended and doubled would raise the cost, and announced does not explain the saving.

5 **Choice A** **Words in context**

One study found the treatment worked and the other that it did not, so the conclusions were opposing. Identical is the reverse.

6 **Choice D** **Text structure and purpose**

The underlined sentence names the addition of pedals as the change that turned a curiosity into transport, which the next sentence confirms with manufacture by the thousand. It does not explain speed or question pedals.

7 **Choice C** **Text structure and purpose**

Marit's inability to remember the gate's colour and her sense of waiting suggest disorientation at change. The passage gives no reason for leaving, no clear anger at the owners and is not about the painting.

8 **Choice B** **Text structure and purpose**

The first half shows the tide erasing names, and the second shows the people walking away lighter. The poem has no holiday, question and answer, or rule with examples.

9 **Choice B** **Cross-text connections**

Text 2 says the studies did not control calories and show only fewer cravings, so they do not show that small meals cut overeating. The dietitian does not endorse or deny cravings entirely, or recommend two meals.

10 **Choice A** **Central ideas and details**

The text says frequent buses raise ridership because riders no longer need timetables, and that frequency can matter as much as speed. It does not claim buses are fastest or that timetables are obsolete.

11 **Choice D** **Central ideas and details**

The text rejects the flowing glass idea and offers the glassmaking method instead. It does not say glass flows, call glaziers unskilled, or describe glass as neither solid nor liquid.

12 **Choice C** **Command of evidence: quantitative**

Fastest growth means the biggest proportional increase, and only the first choice reports this for robotics and compares it with others. The others cite single figures or other clubs.

13 **Choice D** **Command of evidence: quantitative**

The claim is of a quick initial rise and slower later growth, which the first choice reports using the graph. The others describe a fall, steady growth, or only the final value.

14 **Choice A** **Command of evidence: textual**

The claim links her writing to her view, and the quotation describing short essays about single decisions shows her method matching her view that ethics is learned through cases. The others describe her attitude, an example, or her contemporaries.

15 **Choice B** **Command of evidence: textual**

Faster sleep in the cooler room is the direct support for the claim. The other choices describe the design without the result.

16 **Choice C** **Command of evidence: textual**

The text says she cites a miner's 1911 letter that discusses a parliamentary bill in detail. Official histories, voting statistics and earlier accounts are not the evidence described.

17 **Choice C** **Inferences**

Seedlings with varying temperature grew taller and thicker, so moderate variation may stimulate growth. The other choices contradict the result or go beyond it.

Answer explanations

SAT Practice Test 4, Reading and Writing, Module 1

Questions 18 to 33

18 **Choice A** **Inferences**

Visitors lingered when labels asked questions, so the curator likely thinks questions prompt closer looking. The text does not say visitors dislike labels or that facts are useless.

19 **Choice D** **Inferences**

Cafés, galleries and summer visitors outnumbering residents show that tourism has replaced fishing. The other choices contradict the passage.

20 **Choice B** **Form, structure, and sense**

Last summer calls for the simple past tense, discovered. The other forms do not fit the time frame.

21 **Choice A** **Boundaries**

The subject decision is singular, and the phrase beginning with along with does not make it plural, so was agrees.

22 **Choice C** **Boundaries**

Two independent clauses are joined with a semicolon. A comma alone would create a comma splice.

23 **Choice D** **Form, structure, and sense**

The comparison is completed by the clause than I am, so the subject pronoun I is correct.

24 **Choice B** **Form, structure, and sense**

An adverb is needed to modify the verb conducted, so carefully is correct.

25 **Choice A** **Form, structure, and sense**

The introductory phrase beginning with Only after requires inversion in the main clause, so did it conclude is right.

26 **Choice C** **Form, structure, and sense**

The climbers continued despite the hidden summit, so the contrasting conjunction but is correct.

27 **Choice B** **Boundaries**

The possessive relative pronoun whose connects to the discoveries. Who's means who is.

28 **Choice D** **Transitions**

The final plan differs from the first draft, so However is the logical transition.

29 **Choice D** **Transitions**

The sentence contrasts the well-studied honeybee with the poorly known solitary bees, so By contrast is correct.

30 **Choice B** **Transitions**

The studies cast doubt on the mayor's promise, so Yet is the logical transition.

31 **Choice A** **Rhetorical synthesis**

Speed is shown by the short time between the start of construction and opening, which the first choice states. The others give size, a start date or the reason for the bridge.

32 **Choice C** **Rhetorical synthesis**

To introduce the writer, the choice must name her, say what she writes and indicate her standing, which the first choice does.

33 **Choice C** **Rhetorical synthesis**

To explain a cause, the choice needs both the reason and the effect, which the first choice provides. The others state a fact or only the effect.

Answer explanations

SAT Practice Test 4, Reading and Writing, Module 2

Questions 1 to 17

1 **Choice C** **Words in context**

She neither flatters nor condemns and leaves the verdict to readers, which makes her tone dispassionate. Effusive and sentimental imply warmth, and accusatory implies condemnation.

2 **Choice D** **Words in context**

An alliance that dissolved as soon as the common rival was beaten was weak, which tenuous means. Durable contradicts the dissolution.

3 **Choice B** **Words in context**

Ostensibly marks the stated purpose, and the contrast is with the real, underlying purpose. Stated would repeat the surface meaning, and the purpose was not accidental.

4 **Choice A** **Words in context**

Remarks that let everyone hear what they already believed must have been open to many readings, so ambiguous fits. Blunt would be direct, and the others do not explain the effect.

5 **Choice C** **Words in context**

After decades without a fragment, a significant find is remarkable. Predictable is the opposite, and dispensable and tedious do not fit.

6 **Choice A** **Words in context**

An argument that answers every objection and has no gaps is airtight. Porous means full of holes, and fanciful and peripheral do not fit.

7 **Choice D** **Text structure and purpose**

The underlined sentence concedes that the method was tedious and used a single subject but says it still gave the first quantitative description of forgetting. It does not dismiss the work, explain syllable choice, or criticise later researchers.

8 **Choice B** **Text structure and purpose**

The final sentence uses the statue facing the wrong way as an image for a story that credits the wrong person. It says nothing of when the statue was made or of destroying the letter, and the explanation of the land comes earlier.

9 **Choice C** **Text structure and purpose**

The poem begins with what the speaker was told, describes the house's decline, and ends by saying the speaker loves its lean. It contains no construction story, blame or comparison.

10 **Choice B** **Cross-text connections**

Text 2 says earlier hand-copied pamphlets failed to spread and that political support mattered, so the claim overstates the press. The scholar does not deny printing or say earlier reformers avoided pamphlets.

11 **Choice A** **Central ideas and details**

The text says the retreat was plain but the main finding was the faster flow. Satellite photographs are not said to be the only method, the glacier retreated, and meltwater is said to lubricate the bed.

12 **Choice D** **Central ideas and details**

The text contrasts the earlier honour given to copying with the modern idea of originality. It does not say apprentices were forbidden individual styles, that modern painters never copy, or that copying was considered theft.

13 **Choice C** **Command of evidence: quantitative**

A pronounced wet season means a large seasonal swing, which the first choice shows for Altamar compared with the small variation in Brevik. The others describe only Brevik, one month, or a trend common to both.

14 **Choice D** **Command of evidence: quantitative**

The claim is a comparison across age groups with housing highest among the youngest, which the first choice shows. The others omit the comparison or compare only the older groups.

15 **Choice B** **Command of evidence: textual**

The claim concerns warnings that were ignored, which only the final quotation states. The others describe her views on rules, her method or her file.

16 **Choice A** **Command of evidence: textual**

Showing the behaviour at a constant temperature when only the days were shortened isolates day length as the trigger. The other choices describe the design or timing, not the cause.

Answer explanations

SAT Practice Test 4, Reading and Writing, Module 2

Questions 18 to 33

18 Choice B Inferences

Identical weights across different buildings suggest a central authority that set standards. The text does not support abandonment, single-family trade or trade limited to goods sold by weight.

19 Choice D Inferences

A change in the walls' appearance slowed the bees without changing the route, so they were relying partly on visual cues. The other choices are not tested.

20 Choice A Inferences

The board's reversal after the survey suggests it had worried the figures would show a pay gap. The text does not say the board accepted the findings, did not know about comparisons or wished to reduce male pay.

21 Choice B Boundaries

A colon introduces the list that follows a complete clause. A semicolon would require a second independent clause, and a comma or full stop would break the sentence.

22 Choice C Boundaries

The subject of the clause is many, which refers to the plural stories, so were is correct.

23 Choice A Form, structure, and sense

The introductory phrase must modify the team that completed the survey, so the team must be the subject. The other choices make data, the council or a presentation the subject.

24 Choice D Form, structure, and sense

The plural verb work requires a plural antecedent, so those who is correct. Anyone who and whoever would need a singular verb, and whom is the wrong case.

25 Choice B Form, structure, and sense

The subject cheetah is singular, so relies on is correct.

26 Choice C Boundaries

In a conditional sentence with were to, the main clause uses a conditional modal such as could.

27 Choice D Form, structure, and sense

The relative pronoun who refers to the plural scientists, so the verb must be plural: test.

28 Choice A Transitions

The residents' loyalty contrasts with the lack of amenities, so Still is the right transition.

29 Choice C Transitions

The sentence reports what happened later, so Subsequently is the best transition. Accordingly and Likewise do not suit the relation.

30 Choice D Transitions

The second sentence reinforces the first with a fact, so In fact is correct.

31 Choice A Rhetorical synthesis

Rarity is shown by the figure of once every 375 years at a given location, which only the first choice supplies. The others define the event, give a date or describe the visibility.

32 Choice B Rhetorical synthesis

A comparison of use of the form needs both the shared form and the difference in their rhyme, which the first choice does.

33 Choice A Rhetorical synthesis

A finding and a limitation need a result and a restriction on it, which the first choice gives. The others give a result without a limitation or the reverse.

Answer explanations

SAT Practice Test 4, Math, Module 1

Questions 1 to 14

1 **Choice B** **Percentages**

45% of 400 is 0.45 times 400, which is 180.

2 **Choice D** **Data analysis**

Team 1 has 25 and Team 4 has 18, and 25 minus 18 is 7.

3 **Choice C** **Linear equations**

Subtract 3x from both sides: 3x − 11 = 4. Then 3x = 15, so x = 5.

4 **Choice A** **Linear equations in context**

The fee is paid once, so it is the constant 28. The $25 repeats, so it multiplies m. The total is 25m + 28.

5 **Choice A** **Systems of equations**

Subtracting the second equation from the first gives 2y = 20, so y = 10.

6 **Choice D** **Functions**

Each step of 1 in x adds 4 to f(x), so the slope is 4. Then f(1) = 12 gives 4 + 8, so the intercept is 8.

7 **Choice B** **Nonlinear expressions**

Multiply the coefficients, 4 times 5 is 20, and add the exponents, 2 plus 4 is 6.

8 **Choice C** **Triangles**

The angles of a triangle add to 180°, so x + 2x + 6x = 9x = 180 and x = 20.

9 **Choice D** **Circles**

The circumference $2\pi r = 12\pi$ gives r = 6. The area is $\pi r^2 = 36\pi$.

10 **Choice C** **Quadratics**

The equation factors as (x − 8)(x + 2) = 0, so x = 8 or x = −2.

11 **Choice A** **Probability**

38 students travel that way and 17 of them are in Grade 10, so the probability is 17/38. The fraction with 115 on the bottom uses every student.

12 **Choice B** **Statistics**

The first 6 scores total 480, and adding 94 gives 574. Divided by 7, the mean is 82.

13 **Choice D** **Scatterplots**

Substitute x = 7: 7(7) + 54 = 49 + 54 = 103.

14 **Choice A** **Inequalities**

The total is the 36 already joined plus 15 each week, and it must be at least 200, so 36 + 15w ≥ 200.

Answer explanations

SAT Practice Test 4, Math, Module 1

Questions 15 to 27

15 **Choice B** Functions

f(3) = 4(9) − 18 + 6 = 36 − 18 + 6 = 24.

16 **Choice C** Exponential functions

Tripling every 2 hours means the number of triplings is t/2, so the population is 500 times 3 to the power t/2.

17 **Choice D** Linear functions

The slope is (40 − 20)/(7 − 3) = 5. Using (3, 20), 20 = 5(3) + b gives b = 5.

18 **Choice B** Percent change

The increase is 1,800, and dividing by the original 6,000 gives 0.3, which is 30%. Dividing by the new value would give a smaller percent.

19 **Choice C** Volume

The volume is $\pi r^2 h = \pi(36)(9) = 324\pi$.

20 **Choice A** Right triangles

$6^2 + 8^2 = 100$, so the hypotenuse is $\sqrt{100} = 10$.

21 **24** Linear equations

Let the number be n. Then (2/3)n = 16, so n = 16 times 3/2 = 24.

22 **6** Linear equations

Expanding gives 5x − 15 = 8x − 33. Collecting terms gives −3x = −18, so x = 6.

23 **6** Ratios and rates

The car travels 48 miles per gallon, so 288 miles needs 288 / 48 = 6 gallons.

24 **80** Area

The area of a triangle is half the base times the height: (1/2)(16)(10) = 80.

25 **15** Systems of equations

Adding x + y = 26 and x − y = 4 gives 2x = 30, so the larger number is 15.

26 **63** Volume

The volume is $\pi r^2 h = \pi(9)(7) = 63\pi$, so k = 63.

27 **3** Functions

3a − 8 = 1 gives 3a = 9, so a = 3.

Answer explanations

SAT Practice Test 4, Math, Module 2

Questions 1 to 14

1 **Choice C** Linear functions

The slope is (27 − 11)/(7 − 3) = 4. Using (3, 11), 11 = 4(3) + b gives b = −1.

2 **Choice A** Percentages

80% of 200 is 0.8 times 200, which is 160.

3 **Choice B** Probability

30 students travel that way and 18 of them are in Grade 10, so the probability is 18/30, which simplifies to 3/5. The fraction with 111 on the bottom uses every student.

4 **Choice D** Exponential functions

Tripling every 4 hours means the number of triplings is t/4, so the population is 500 times 3 to the power t/4.

5 **Choice B** Quadratics

The equation factors as (x − 2)(x + 4) = 0, so x = 2 or x = −4.

6 **Choice C** Volume

The volume is $\pi r^2 h = \pi(25)(10) = 250\pi$.

7 **Choice A** Inequalities

The total is the 48 already joined plus 20 each week, and it must be at least 150, so 48 + 20w ≥ 150.

8 **Choice D** Exponential functions

A year is 12 months, so the value is multiplied by 0.7^(12/6) = 0.7^2 = 0.49 each year. That is a decrease of 1 − 0.49, or 51%.

9 **Choice D** Circles

Arc length is the radius times the central angle in radians: 9 × π/3 = 3π.

10 **Choice C** Equivalent expressions

$(4x + 3)^2 = 16x^2 + 24x + 9$, and $(3x − 1)^2 = 9x^2 − 6x + 1$. Subtracting the second from the first gives $7x^2 + 30x + 8$.

11 **Choice A** Percentages

Start at 100. After the 25% rise the price is 125, and 20% off 125 leaves 100. That is 0% higher than the original.

12 **Choice B** Absolute value

The inequality means x is within 4 of 5, so 1 ≤ x ≤ 9. That range has 9 integers, counting both ends.

13 **Choice B** Exponential equations

Write 4 as 2^2, so the right side is 2^(2x − 8). Equal bases give x + 5 = 2x − 8, so x = 13.

14 **Choice A** Circles in the plane

The center of circle A is (4, −1). Shifting it 3 units left and 5 units down moves the center to (1, −6), and the radius is unchanged, so the equation is (x − 1)2 + (y + 6)2 = 64.

Answer explanations

SAT Practice Test 4, Math, Module 2

Questions 15 to 27

15 **Choice D** Quadratics

Completing the square: $x^2 - 8x + 9 = (x - 4)^2 - 7$. The smallest value of the square is 0, so the minimum of f is −7.

16 **Choice C** Percentages

180 is 150 more than 30, and 150 / 30 = 5, so p = 500.

17 **Choice D** Functions

g(2) = 8, so f(g(2)) = f(8) = 36. Also f(2) = 12, so g(f(2)) = g(12) = 148. The difference is 36 − 148 = −112.

18 **Choice B** Trigonometry

A right triangle with opposite side 8 and hypotenuse 17 has adjacent side 15, so tan A = opposite/adjacent = 8/15.

19 **Choice C** Ratios and rates

The pumps fill 1/12 and 1/6 of the tank each hour, together 1/4 of it, so the time is 4 hours.

20 **Choice A** Equivalent expressions

The numerator factors as (x − 4)(x − 6). Cancelling x − 4 leaves x − 6.

21 **26** Systems of equations

Adding x + y = 35 and x − y = 17 gives 2x = 52, so the larger number is 26.

22 **11** Quadratics

For $ax^2 + bx + c = 0$ the solutions add to $-b/a = -(-22)/2 = 11$.

23 **20** Percentages

Let L be the litres of the 20% solution. Then 0.2L + 0.8(10) = 0.4(L + 10). Solving gives L = 20.

24 **14** Systems of equations

Adding the equations gives 5x + 5y = 70. Since the coefficients are equal, 5(x + y) = 70, so x + y = 14.

25 **45** Similar triangles

The scale factor is DE / AB = 9 / 3 = 3. The perimeter of ABC is 15, so the perimeter of DEF is 3 × 15 = 45.

26 **10** Quadratics

Exactly one real solution means the discriminant is zero: $b^2 - 4(25) = 0$, so $b^2 = 100$ and b = 10 (taking the positive value).

27 **-5** Systems with nonlinear equations

Setting the expressions equal gives $x^2 - 6x + (4 - k) = 0$. One solution means the discriminant is 0: 36 − 4(4 − k) = 0, so k = −5.